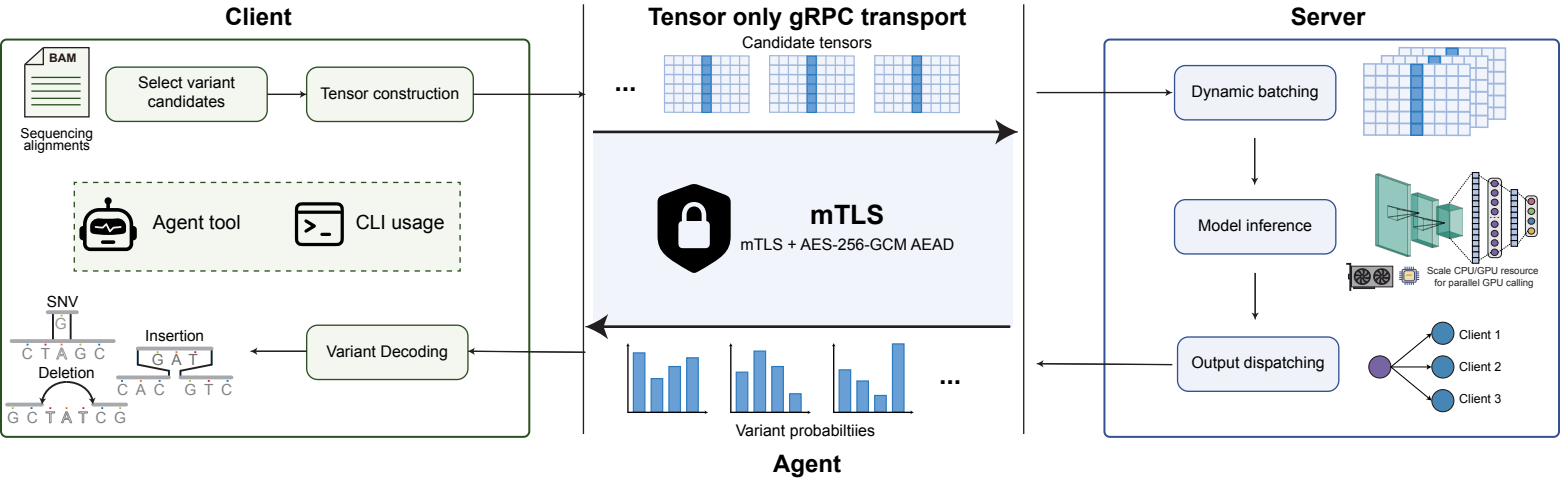
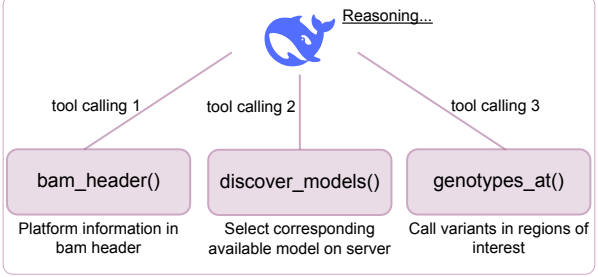



## Agent

### 1) Task prompt

```
Clinical genomics. Sample HG00136 is a human whole-genome
sequencing dataset registered with the service. Determine its
APOE diplotype — the epsilon2/epsilon3/epsilon4 combination
that stratifies Alzheimer's-disease risk — defined by two SNPs
on chr19 (GRCh38): rs429358 at chr19:44908684 (ref T, alt C)
and rs7412 at chr19:44908822 (ref C, alt T).
The three common haplotypes are:
epsilon4 = rs429358 C, rs7412 C
epsilon3 = rs429358 T, rs7412 C
epsilon2 = rs429358 T, rs7412 T
Steps:
1. Determine which sequencing platform this sample is and
select the matching Clair3 model.
2. Call variants and read the genotype at both SNP sites. A
site with no called variant is homozygous reference.
3. Combine the two genotypes into the standard clinical
diplotype.
Finish with one line:
ANSWER: <diplotype>
For example:
ANSWER: e4/e4
```

### 2) LLM agent (Deepseek-v4-pro)




### 3) APOE interpretation

